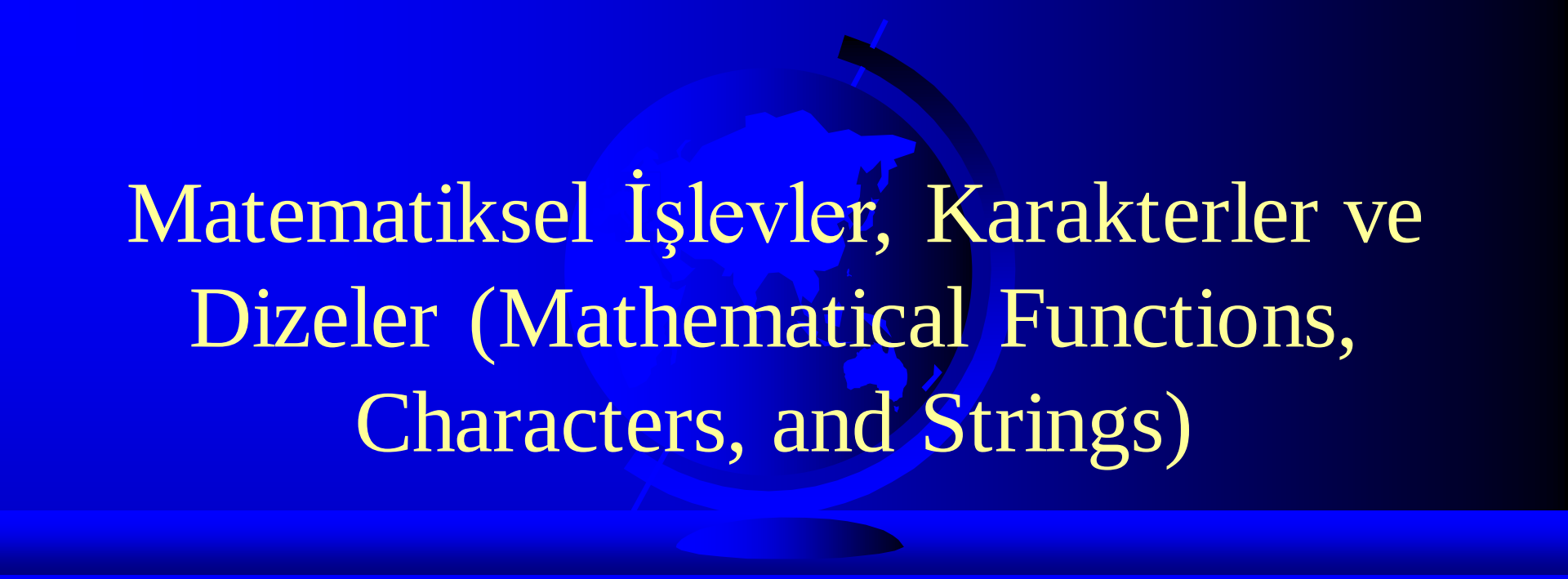


Matematiksel İşlevler, Karakterler ve Dizeler (Mathematical Functions, Characters, and Strings)

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Matematiksel Fonksiyonlar

Java, Math sınıfında, ortak matematiksel işlevleri yerine getirmek için birçok yararlı yöntem sunmaktadır.

Math sınıfı iki faydalı double sabit (constant) sağlar: PI ve E (doğal logaritmaların tabanı). Herhangi bir programda bu sabitleri Math.PI ve Math.E olarak kullanabilirsiniz.

Trigonometrik Yöntemler

Method	*Description*
`sin(radians)`	Returns the trigonometric sine of an angle in radians.
`cos(radians)`	Returns the trigonometric cosine of an angle in radians.
`tan(radians)`	Returns the trigonometric tangent of an angle in radians.
`toRadians(degree)`	Returns the angle in radians for the angle in degrees.
`toDegrees(radians)`	Returns the angle in degrees for the angle in radians.
`asin(a)`	Returns the angle in radians for the inverse of sine.
`acos(a)`	Returns the angle in radians for the inverse of cosine.
`atan(a)`	Returns the angle in radians for the inverse of tangent.

Math.toDegrees(Math.PI / 2)returns90.0

Math.toRadians(30) returns 0.5236 (same as π/6)

Math.sin(0) returns 0.0

Math.sin(Math.toRadians(270)) returns −1.0

Math.sin(Math.PI / 6) returns 0.5

Math.sin(Math.PI / 2) returns 1.0

Math.cos(0) returns 1.0

Math.cos(Math.PI / 6) returns 0.866

Math.cos(Math.PI / 2) returns0

Math.asin(0.5) returns 0.523598333 (same as π/6)

Math.acos(0.5) returns 1.0472 (same as π/3)

Math.atan(1.0) returns 0.785398 (same as π/4)



Üs AlmaYöntemleri (Exponent Methods)

Method	*Description*
`exp(x)`	Returns e raised to power of x (e^x).
`log(x)`	Returns the natural logarithm of x ($\ln(x) = \log_e(x)$).
`log10(x)`	Returns the base 10 logarithm of x ($\log_{10}(x)$).
`pow(a, b)`	Returns a raised to the power of b (a^b).
`sqrt(x)`	Returns the square root of x ($\sqrt{x}$) for $x >= 0$.

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Üs Alma İşlemleri (Exponent Operations)

- Math.pow(**2**, **3**), which returns **8.0**
- Math.exp(**3.5**), which returns **33.11545**
- Math.sqrt(**4**), which returns **2.0**
- Math.log(Math.E), which returns **1.0**



Yuvarlama Yöntemleri (The Rounding Methods)

Method	*Description*
`ceil(x)`	*x* is rounded up to its nearest integer. This integer is returned as a double value.
`floor(x)`	*x* is rounded down to its nearest integer. This integer is returned as a double value.
`rint(x)`	*x* is rounded to its nearest integer. If *x* is equally close to two integers, the even one is returned as a double value.
`round(x)`	Returns `(int)Math.floor(x + 0.5)` if *x* is a float and returns `(long)Math.floor(x + 0.5)` if *x* is a double.

Math.ceil(**2.1**) , returns **3.0**
Math.floor(**2.1**) , returns **2.0**
Math.rint(**−2.1**) , returns **−2.0**
Math.round(**2.6f**) , returns **3** // Returns int
Math.round(**2.0**) , returns **2** // Returns long



Minimum, Maksimum ve Mutlak Değer Yöntemleri (The **min**, **max**, and **abs** Methods)

- Abs yöntemi sayının mutlak değerini döndürür (int, long, float veya double)

Math.max(**2**, **3**) returns **3**
Math.min(**2.5**, **4.6**) returns **2.5**
Math.max(Math.max(**2.5**, **4.6**), Math.min(**3**, **5.6**)) returns **4.6**
Math.abs(**−2**) returns **2**
Math.abs(**−2.1**) returns **2.1**



Rastgele Yöntemi (The **random** Method)

Bu yöntem, 0.0'a eşit veya daha büyük ve 1.0'dan küçük bir rasgele çift değer üretir.(**0 <= Math.random() < 1.0**).

(int)(Math.random() * **10**) // 0 ile 9 arasında rastgele bir tamsayı döndürür.

a + Math.random() * b // a + b hariç, a ile a + b arasında rasgele bir sayı döndürür.



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Karakter Veri Tipi ve İşlemleri

- Karakter veri türü (char), tek bir karakteri temsil etmek için kullanılır. Bir karakter değişmezi tek tırnak işareti içine alınır.

- **char** letter = '**A**';
- **char** numChar = '**4**';



```
char letter = 'A';
char letter = '\u0041'; // Character A's Unicode is 0041
```

Characters	*Code Value in Decimal*	*Unicode Value*
'0' to '9'	48 to 57	\u0030 to \u0039
'A' to 'Z'	65 to 90	\u0041 to \u005A
'a' to 'z'	97 to 122	\u0061 to \u007A



Karakterleri Karşılaştırma ve Test Etme

'a' < 'b' is true because the Unicode for **'a'** (**97**) is less than the Unicode for **'b'** (**98**).

'a' < 'A' is false because the Unicode for **'a'** (**97**) is greater than the Unicode for **'A'** (**65**).

'1' < '8' is true because the Unicode for **'1'** (**49**) is less than the Unicode for **'8'** (**56**).



Karakter Sınıfındaki Yöntemler

Character sınıfı, java.lang paketinde tanımlanmıştır. (The **Character** class is defined in the **java.lang** package)

Method	*Description*
`isDigit(ch)`	Returns true if the specified character is a digit.
`isLetter(ch)`	Returns true if the specified character is a letter.
`isLetterOrDigit(ch)`	Returns true if the specified character is a letter or digit.
`isLowerCase(ch)`	Returns true if the specified character is a lowercase letter.
`isUpperCase(ch)`	Returns true if the specified character is an uppercase letter.
`toLowerCase(ch)`	Returns the lowercase of the specified character.
`toUpperCase(ch)`	Returns the uppercase of the specified character.

isDigit('a') is false

isLetter('a') is true

isLowerCase('a') is true

isUpperCase('a') is false

toLowerCase('T') is t

toUpperCase('q') is Q



String Tipi

String, bir karakter dizesidir.

String message = **"Welcome to Java"**;



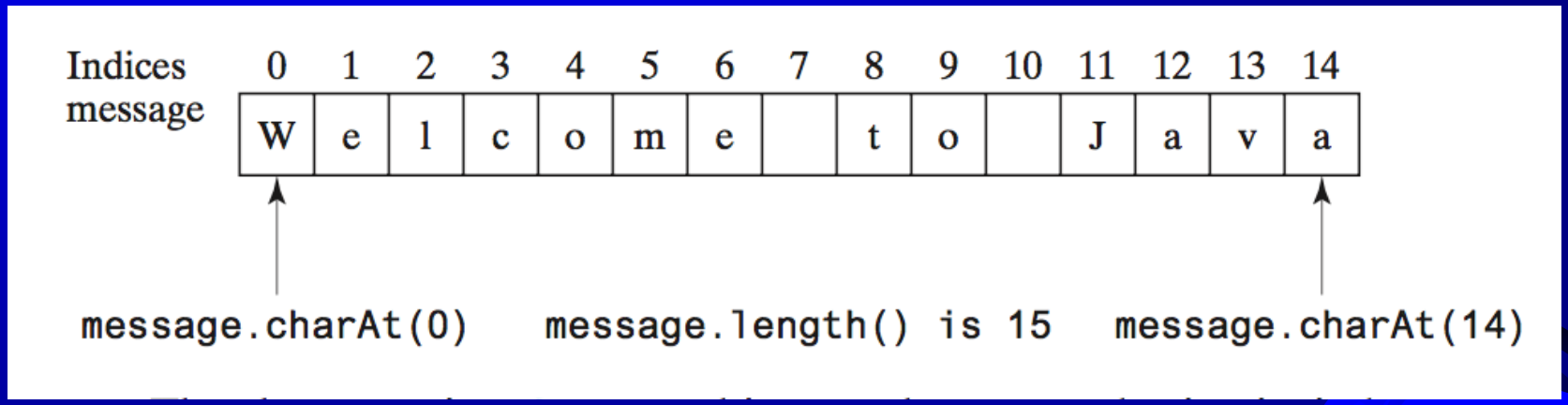
String Nesneleri için Basit Yöntemler

Method	*Description*
`length()`	Returns the number of characters in this string.
`charAt(index)`	Returns the character at the specified index from this string.
`concat(s1)`	Returns a new string that concatenates this string with string `s1`.
`toUpperCase()`	Returns a new string with all letters in uppercase.
`toLowerCase()`	Returns a new string with all letters in lowercase.
`trim()`	Returns a new string with whitespace characters trimmed on both sides.



String' den Karakter Almak

S.charAt (index) yöntemi, s dizesinde belirli bir karakteri almak için kullanılabilir; burada dizin, 0 ile s.length () - 1 arasındadır.



```
String message = "Welcome to Java";
System.out.println("The length of " + message + " is "
+ message.length());

System.out.println( "Welcome".toLowerCase()); //returns
welcome

System.out.println( "Welcome".toUpperCase()); //returns a new
string WELCOME.

System.out.println( message.charAt(0)); //returns the character
W
```

